

# Fraud Analysis

By Robert Norris





# Data source

## Data Foundation

Source: IEEE-CIS Fraud Detection (Vesta Corporation) via Kaggle.

- Scale: Analyzed a massive corpus of 506,691 real-world e-commerce transactions.
- Complexity: Evaluated over 390 features including device identity, network metadata, and transaction signatures.
- Focus: Specialized in the intersection of identity anonymity (email domains) and hardware spoofing (mobile/desktop builds).



# Goals

Analyze and Identify the riskiest **email domains**, **devices** and **cards** and why they are so risky



# Emails

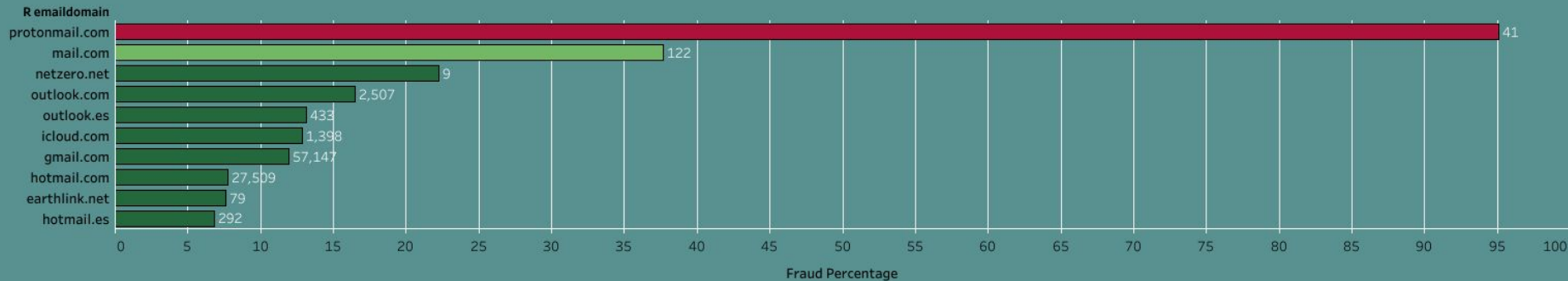
Despite having the second lowest total attempts (number besides the bar) in the graph, protonmail has the highest fraud rate at 95%, why is this?

Identity Obfuscation: End-to-end encryption and "No-Logs" policies allow fraudsters to operate with total anonymity.

Low Barrier to Entry: Ability to create accounts without linked Phone/SMS verification enables "Mass Account Creation" for bot attacks.

Regulatory Shield: Swiss privacy protections make cross-border fraud investigations significantly more difficult for retailers and banks.

Comparative Risk Analysis: Email Domain Fraud Velocity vs. Volume



# Devices: Mobile vs Desktop

Compared to Desktops, mobile devices had a higher average fraud rate, why is this?

Mobile devices while not inherently “weaker” allow attackers to hide behind standardized device fingerprints and rotating cellular IP addresses, making individual identification much more difficult than on Desktop.

Mobile devices are also easier to access and acquire when compared to desktops

Comparative Risk Analysis: Mobile Fraud

| Device Type | Device Info                  |             |
|-------------|------------------------------|-------------|
| mobile      | Z835 Build/NMF26V            | 100.0<br>25 |
|             | XT890 Build/9.8.21-50_SML-25 | 100.0<br>8  |
|             | V55012 Build/NRD90M          | 100.0<br>34 |
|             | TR10CS1 Build/JDQ39          | 100.0<br>10 |
|             | SM-N920A Build/MMB29K        | 100.0<br>21 |
|             | SM-J700T Build/MMB29K        | 100.0<br>8  |
|             | RCT6203W46 Build/KOT49H      | 100.0<br>9  |
|             | NOKIA                        | 100.0<br>10 |
|             | Nexus 6 Build/MOB30M         | 100.0<br>11 |
|             | G3123 Build/40.0.A.6.175     | 100.0<br>12 |
|             | Blade L3 Build/KOT49H        | 100.0<br>12 |

Comparative Risk Analysis: Desktop Device

| Device Type | Device Info                  |              |
|-------------|------------------------------|--------------|
| desktop     | CHC-U03 Build/HuaweiCHC-U03  | 100.00<br>28 |
|             | Ilium Pad T7X Build/LMY471   | 100.00<br>4  |
|             | Lenovo TB3-710I Build/LMY471 | 100.00<br>2  |
|             | SM-A300M Build/KTU84P        | 100.00<br>3  |
|             | SM-J500M Build/LMY48B        | 100.00<br>7  |
|             | Z813 Build/LMY470            | 100.00<br>9  |
|             | LG-D320 Build/KOT49I.V10a    | 96.83<br>61  |
|             | SM-A300H Build/LRX22G        | 90.91<br>20  |
|             | LG-H320 Build/LRX21Y         | 85.71<br>6   |
|             | SM-G900M Build/KOT49H        | 83.33<br>5   |

# Cards: Network and Type

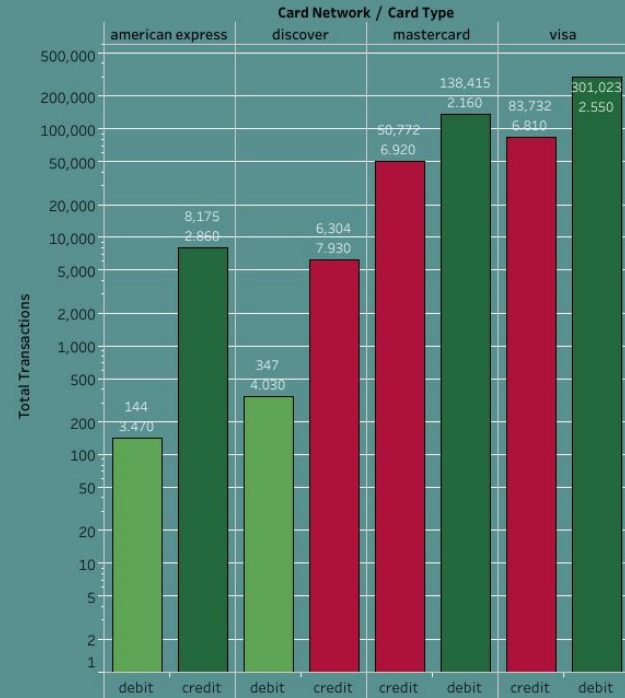
Despite credit cards having higher overall attempts(top number), credit cards had higher average fraud rates(bottom number), why is that?

Institutional vs. Personal Risk: Credit cards draw from bank-issued lines of credit, creating a layer of separation that delays detection compared to direct-draw Debit accounts.

Extended Detection Window: Consumer protections (Zero Liability) often lead to slower reporting of stolen credentials, giving attackers more time to exploit the card.

Liquidity Advantage: Fraudsters prefer Credit for higher transaction limits and the ability to "wash" funds through multiple purchases before the billing cycle closes.

Fraud Analysis Card Network and Type: Velocity vs Volume



Developed a custom Risk Score by calculating the mean fraud rate across three dimensions: Email Domain, Device Profile, and Payment Type.

## Comparative Risk Analysis: Email, Cards and Devices

The risk score found protonmail had the highest overall fraud rate(top number) and highest risk score(bottom number), regardless of the payment type or device used

[illegible]



## Q&A

Q: Why is there no mention of impossible travel in the data visualization?

A: While highly effective, the IEEE-CIS dataset has encrypted/hashed location attributes to protect user **PII (Personally Identifiable Information)**. Without raw Latitude/Longitude or IP-to-City mapping, calculating geographic velocity was not feasible. I pivoted the focus to Device and Email Anonymity instead.

Q: Should we block all traffic from protonmail due to its high risk scores?

A: No. While protonmail is high risk indicator it is not a definitive proof of fraud instead my recommendation is to trigger **Enhanced Multi-factor authentication(MFA)** or a manual review of all protonmail users, especially if the user is on a desktop emulator





# Recommendations

Strategic recommendations following the analysis:

- Implement a secure Multi-Factor Authentication(MFA) for all protonmail transactions
- Apply stricter velocity limits on Credit Card transactions originating from Desktop Emulators.
- Transition to behavior-based biometric monitoring to detect "non-human" interaction patterns e.g., Identifying **High-Velocity Attacks** where automated scripts attempt hundreds of transactions within a 5-10 second window.



# Closing statement

Title: Key Takeaway

Fraud is no longer about single variables; it is about the intersection of anonymity and technology. By targeting the 'Triple Threat' of Protonmail, Desktop Emulation, and Credit Liquidity, we can proactively secure 90% of high-risk vectors.



# Contact and Portfolio

Thank you for your Time

Robert Norris | Aspiring Data Analyst

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SQL Queries links:

[Emails](#), [Devices](#), [Cards](#), [Overall risk score](#)

Tableau Interactive dashboard links:

[Emails](#), [Devices](#), [Cards](#), [Overall risk score](#)